SECURITY ASSESSMENT

Grand Marina Hotel Water Monitoring System

Prepared for: Marcus Webb, General Manager

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EXECUTIVE SUMMARY

Ultimately, Marcus should deploy security for the hotel's monitoring system immediately. As of now, all sensor data, including device IDs, time stamps and pressure readings, are sent across the network in plain text. Meaning anyone that is on the hotel WI-Fi can read it without them trying that hard.

By adding TLS encryption and device authentication, it fixes that. Data becomes scrambled in transit, and only approved devices can connect. The system still handles about 97 sensor messages per second with each reading arriving every 0.05 seconds, which is pretty much instantaneous.

Hotels have been hit by IoT breaches before, specifically where unsecured devices exposed operational and guest information. Industry leaders such as IBM, Palo Alto Networks and Microsoft all stress that encrypting IoT traffic is very essential. Based on our tests and research within real- world environments, deploying security now is safe, fast and necessary to keep everything running smoothly.

SECURITY RISKS ADDRESSED

Before Security:

Every sensor reading travels openly across the hotel network, which is the equivalent of sending sensitive information on a postcard! During testing, we could see device ID’s, timestamps, and pressure readings instantly. These risks include, anyone that is on the network can monitor hotel operations, malicious actors could track sensor patterns or interfere with devices, and guests privacy could be violated putting the hotels reputation at risk.

After Security:

With encryption in place, that data is no longer readable to outsiders. If someone tries to intercept it, they either see an error with nothing or only encrypted data that they cannot understand. Also, if someone tries to connect without permission, they will get blocked. Overall, it makes the system a lot more secure and protects both the hotel and its guests.

PERFORMANCE IMPACT

What We Measured	Before Security	After Security	Verdict
How fast readings arrive	[0.05] seconds	[0.05] seconds	No noticeable difference
How many sensors at once	[96.8] per sec	[96.8] per sec	Meets requirements

The system handles almost the full load with no errors, and security does not really slow it down. Everything still works just as smooth, so there's no downside to keeping it secure. Specifically, each sensor takes 0.05 seconds to arrive. This is effectively instantaneous for hotel operations, so security does not slow down day-to-day monitoring. The system can handle about 97 sensor messages per second , which is more than enough to suppose all sensors report at once. Even under peak stress, there were no delays or errors.

```
==================================================
  Stress Test Results (TLS OFF)
==================================================
  Target rate: 100 msg/sec
  Actual rate: 96.8 msg/sec
  Messages sent: 3000
  Errors: 0
  Success rate: 100.0%
  Status: SUCCESS (achieved target rate)
==================================================
```

```
==================================================
  Stress Test Results (TLS ON)
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TESTING EVIDENCE

We ran four security tests on the water monitoring system:

1. Can outsiders see sensor data?

Before security: Everything was visible in plain text. Outsiders could easily read device information, timestamps, and readings.

After security was added: No data could be read. It was either blocked or completely scrambled, so outsiders couldn't understand anything.

```
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % mosquitto_sub -h localhost -p 1883 -t "grandmarina/#" -v
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 1, "pressure_psi": 82.5, "flow_rate_gpm": 45.0, "timestamp": "14
:39:00"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 2, "pressure_psi": 83.5, "flow_rate_gpm": 45.0, "timestamp": "14
:39:01"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 3, "pressure_psi": 84.5, "flow_rate_gpm": 45.0, "timestamp": "14
:39:02"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 4, "pressure_psi": 85.5, "flow_rate_gpm": 45.0, "timestamp": "14
:39:03"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 5, "pressure_psi": 86.5, "flow_rate_gpm": 45.0, "timestamp": "14
:39:04"}
^C%                                                         kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % mosquitto_s
ub -h localhost -p 8883 -t "grandmarina/#" -v

Error: Protocol error
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project %
```

2. Does the system block unauthorized connections?

When a device tries to connect without the correct credentials, the connection fails and shows an error. Only devices with the correct setups are able to connect successfully.

```
Done publishing!
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode connect --tls on

==================================================
  Connection Test
  TLS: ON
  CA Certificate: certs/ca.pem
==================================================

SUCCESS: Connected to broker!
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode generate-wrong-ca


Generating wrong CA certificate for testing...
Saved: certs/wrong-ca.pem
To test: Use this as ca_certs in your client
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode test-wrong-ca
Testing connection with wrong CA...

==================================================
  Connection Test
  TLS: ON
  CA Certificate: certs/wrong-ca.pem
==================================================

FAILED: [SSL: CERTIFICATE_VERIFY_FAILED] certificate verify failed: self-signed certificate in certificate chain (_ssl.c:1081)
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode connect --tls on --no-ca

==================================================
  Connection Test
  TLS: ON
  CA Certificate: NONE
==================================================

SUCCESS: Connected to broker!
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project %
```

3. Does security slow things down?

Not really. The delay added by encryption was pretty much un-noticable. The difference is only a few milliseconds, which is so small for anyone to actually notice within real use.

```
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode latency --tls off --count 50

==================================================
  Latency Test
  TLS: OFF
  Messages: 50
==================================================

  Sent 10/50 messages...
  Sent 20/50 messages...
  Sent 30/50 messages...
  Sent 40/50 messages...
  Sent 50/50 messages...

==================================================
  Latency Results (TLS OFF)
==================================================
  Messages sent: 50
  Average latency: 52.75 ms
  Min latency: 0.65 ms
  Max latency: 105.80 ms
  Std deviation: 51.72 ms
==================================================

kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode latency --tls on --count 50


==================================================
  Latency Test
  TLS: ON
  Messages: 50
==================================================

  Sent 10/50 messages...
  Sent 20/50 messages...
  Sent 30/50 messages...
  Sent 40/50 messages...
  Sent 50/50 messages...

==================================================
  Latency Results (TLS ON)
==================================================
  Messages sent: 50
  Average latency: 52.55 ms
  Min latency: 0.56 ms
  Max latency: 106.17 ms
  Std deviation: 51.55 ms
==================================================

kaylacolletti@Kaylas-MacBook-Pro hydroficient-project %
```

4. Can the system handle a crisis?

Yes, the system handled normal, peak and even high-stress conditions without failing. It reached about 96.8 messages per second with no errors, which meets the requirements. Even under pressure, everything kept running smoothly.

```
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RECOMMENDATION

Security must be added to the system. Without it, anyone on the network could see all the data and possibly mess with the system, which is a big risk for both operations and guest privacy. Companies such as IBM and Palo Alto Networks explain that unprotected devices can expose sensitive information and allow unauthorized access. After adding security, the data is no longer readable and only approved devices can connect. Microsoft also emphasizes that encryption helps protect data and keep systems secure. Based on our tests, security did not slow anything down, so there are no downsides and it is definitely worth it to keep everything protected.

NEXT STEPS

1. Turn on security now: Set up encryption across the system (TLS encryption active, data scrambled, unapproved devices blocked). Will see that the data is no longer readable and random devices cannot connect.
2. Only allow approved devices: Make sure every device needs the correct credentials to connect. This stops unauthorized access completely.
3. Keep monitoring the system: Check for any unusual activity and keep everything updated. This helps catch problems early and ensures the system's security over time.
4. Document everything: Keep a record for hotel management and future audits.

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